LAB 5: Object Oriented Programming in C++

Goal

- Understand the basic concept of classes
- Implement own defined C++ class
- Understand the concept of inheritance
- Understand the concept of polymorphism

Task preparation

There are six tasks in this lab. No submission required before the lab. Codes will be checked by lab assistants during the lab session.

Lab tasks:

Task 1

Execute the code examples from Lecture 9 code_6 to code_20 mentioned in the lecture slides. The code examples for can be found in Lisam>Course documents>Lectures.

No need to show anything in this task, the goal is to make sure that there are no questions about the code introduced during the lecture.

Task 2

Write a program using classes to compute hypotenuse of triangle where the object attributes contain lengths and a method to compute the hypotenuse.

Class must be defined as follows:

- Two private variables length x and y
- One method to compute hypotenuse of triangle with length of other sides x and y

  hypotenuse $= \sqrt{x^2 + y^2}$
- get() and set() helper functions to access private variables
- Use `#include <cmath>` header for sqrt() and pow() functions

Task 3

Write a program where base class employee with employee name (string), id (integer), and salary (integer) will be inherited by derived class employee_function with variables for department (string type), manager or not (Boolean type)

Task 4

Write a program to define the following base class and derived class that inherits from base class

- Base class: employee, public variables: employee name (string), id (integer)
- Base class: employee, private variable salary (integer)
- Derived class: employee_task, public variable: department (string type)
- Derived class employee_task, private variable: manager or not (Boolean type)

Task 4.1:

- Set access level protected for inheritance
- Create object in main()
- Set and display values for the objects

Task 4.2:

- Set access level private for inheritance
- Create object in main()
- Set and display values for the objects

Task 5

Complex number: The following code is the class header describing a complex number class.

```
#ifndef _COMPLEXH
#define _COMPLEXH
#include<iostream>

class Complex {
public:
Complex(); //Prototype. You should define this constructor.
Complex(double re, double im); //Prototype. You must define this constructor
~Complex(); //Prototype. You should define this destructor.
double getReal(void); //get the real part of the complex number
double getImaginary(void); //get the Imaginary part of the



//complex number
void setReal(double re);//set the real part of the complex number
void setImaginary(double im); //set the Imaginary part of the
//complex number

//overwrite the operators for complex numbers
Complex operator+(Complex& theComplex);
Complex operator-(Complex& theComplex);
Complex operator*(Complex& theComplex);
Complex operator/(Complex& theComplex);
private:
double real;
double imag;
};
#endif
```

- Create a project and name the project as "complex".
- Create a header file called "complex.h" in the header file folder.
- Create a resource file called "complex.cpp" which implements the functions and overloaded operators declared in the complex.h header file.
- Note in the definition of the operators(+,-,*,/), a concept named "reference" is mentioned during the lecture, please learn more about "reference" by yourself when implementing these functions following this link: http://en.wikipedia.org/wiki/Reference_(C++)
- Show the calculation result of the complex number in the main function.

Task 6

Write a program to exchange values of variables of two objects of same class using pointers.

- Create two objects of the same class
- Assign values to variables of the objects
- Exchange the value of the variables using pointers